OSI etalon modeli va TCP/IP modeli

OSI Reference Model and TCP/IP Model

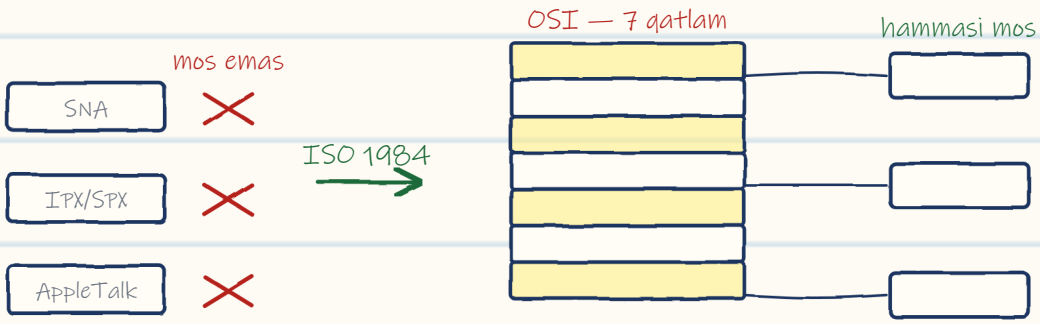
Asosiy atamalar

- OSI modeli — ISO ning 7 qatlamli etalon modeli (1984)
- Qatlam — aniq bir vazifani bajaradigan mustaqil bosqich
- PDU — qatlamning ma'lumot birligi: segment · paket · kadr · bit
- Sarlavha (header) — qatlam qo'shgan xizmat maydonlari
- Enkapsulyatsiya — har qatlam o'z sarlavhasini qo'shishi
- TCP/IP modeli — amalda ishlatiladigan 4 qatlamli model

1. Nega model kerak?

Har vendor o'z protokolini yozdi — qurilmalar bir-biri bilan ishlamadi.

ISO 1984-yilda OSI etalon modelini chiqardi.



1-rasm. Tarqoq standartlar → bitta umumiy model

2. Qatlamli tuzilma nima beradi

- ochiq, standart interfeyslar
- vendorlar orasida moslik
- qatlamlar mustaqil — o'rganish va yangilash oson
- modulli ishlab chiqish, xarajat kam
- nosozlikni qatlam bo'yicha topish

3. Yettita qatlam

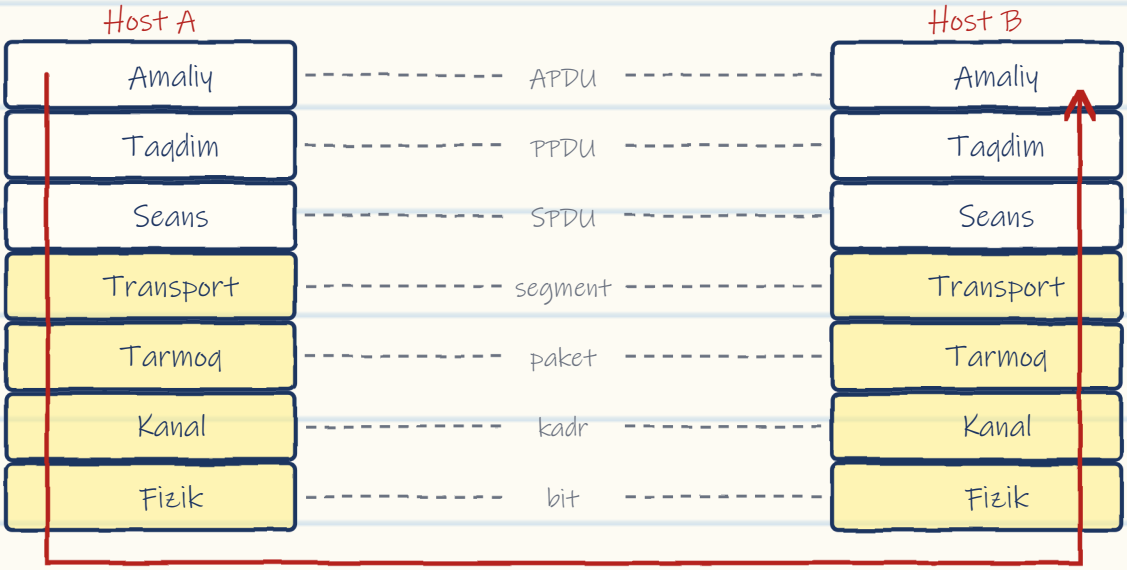
N	qatlam	vazifa	PDU
7	Amaliy	ilovalar aloqasi	APDU
6	Taqdim	format · shifrlash	PPDU
5	Seans	seansni boshqarish	SPDU
4	Transport	uchdan-uchga ulanish	segment
3	Tarmoq	adres · marshrut tanlash	paket
2	Kanal	muhitga kirish · link	kadr
1	Fizik	bit oqimini uzatish	bit

2-rasm. Qatlam · vazifa · ma'lumot birligi

- 7–5 — ilova qatlamlari (yuqori)
- 4–1 — uzatish qatlamlari (pastki)

> Qoida: har qatlam faqat qo'shni qatlam bilan almashadi.

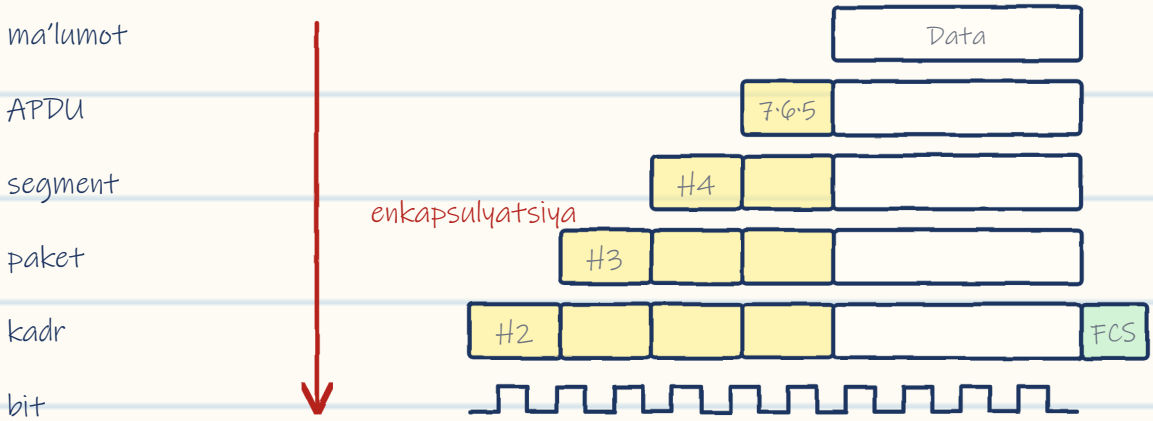
4. Teng qatlamlar aloqasi



3-rasm. Punktir — mantiqiy aloqa, qizil — haqiqiy yo'l

5. Enkapsulyatsiya

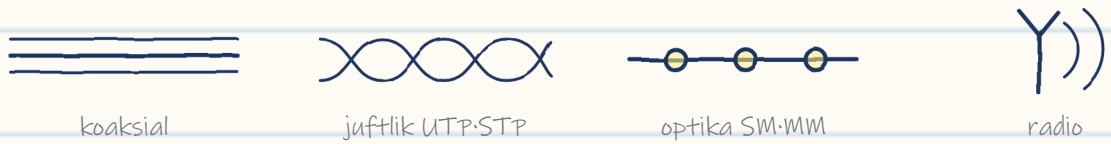
Har qatlam **o'z sarlavhasini qo'shadi**; qabul qilishda teskari.



4-rasm. Sarlavhalar ketma-ket qo'shiladi, data — foydali yuk

6. Fizik qatlam

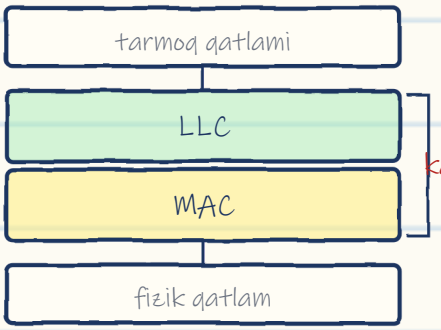
mexanik · elektr · funksional · protsedura



LAN fizik	WAN fizik
10Base-T · 100Base-TX/FX 1000Base-T/SX · hub, repeater	RS-232 · V.24 · V.35 modem

5-rasm. Muhit, standart va qurilmalar

7. Kanal qatlami

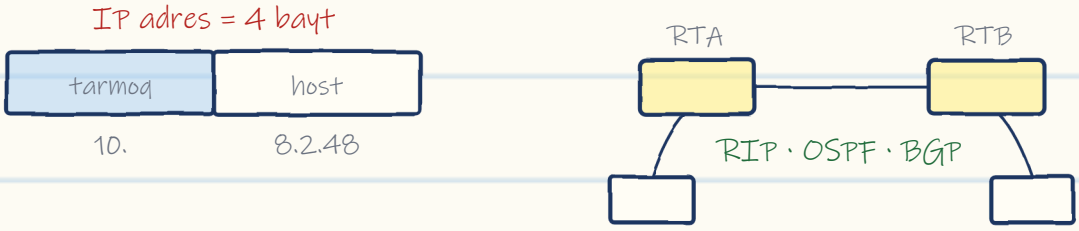


802.2 — LLC
802.3 Ethernet
802.4 token bus
802.5 token ring

kadr · xato nazorati · MAC adres · oqim nazorati

LLC	802.2 — yuqori qatlam protokolini belgilaydi
MAC	muhitga kirish · MAC adres · FCS
WAN	HDLC · PPP · X.25 · Frame Relay

8. Tarmoq qatlami



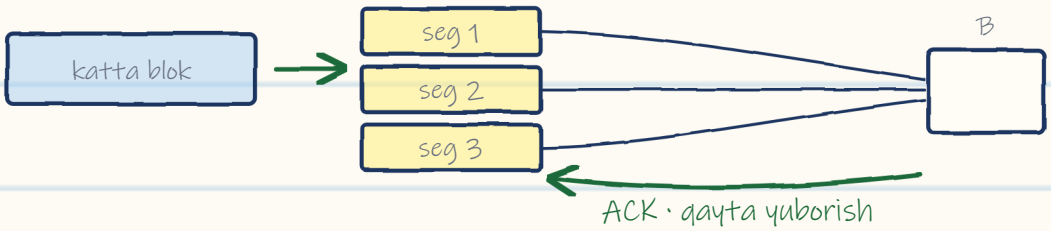
routed = IP · routing = RIP/OSPF/BGP

adreslash · marshrut · tirbandlik · MTU

OSI tarmoq qatlami — ulanishsiz xizmat

6-rasm. Adres = tarmoq + host; marshrut jadval orqali

9. Transport qatlami

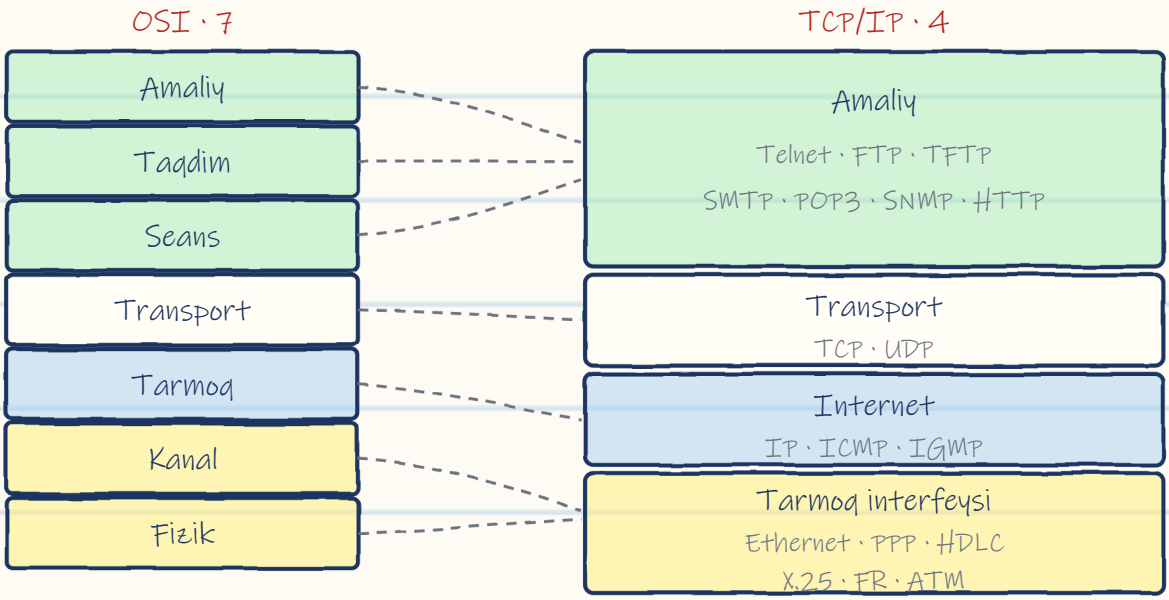


segmentlash · tartib raqami · oqim nazorati

TCP — ishonchli · UDP — ishonchsiz

	ulanishli	ulanishsiz
ulanish	oldin o'rnatiladi	kerak emas
tartib, ACK, qayta yuborish	bor	yo'q

10. TCP/IP modeli



Xulosa

- OSI — ISO, 1984, nazariy 7 qatlam
- PDU: APDU · PPDU · SPDU · segment · paket · kadr · bit
- Enkapsulyatsiya — har qatlam sarlavha qo'shadi
- TCP/IP — amalda ishlatiladigan 4 qatlam
- 5+6+7 → Amaliy; 1+2 → Tarmoq interfeysi

Tekshirish

1. 4-qatlam PDU nomi? — segment
2. LLC standarti? — IEEE 802.2
3. TCP/IP da Internet qatlami protokollari? — IP, ICMP, IGMP
4. V.35 maksimal tezligi? — 2 Mbit/s